

Red-Teaming Privacy-Protective Perturbations: Blind Face Restoration as an Attack Strategy

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Abstract

The rapid growth of individual image generation on online platforms enables users to create personalized visual content on the Web, but also raises serious concerns about users' privacy and portrait rights. A malicious forgery attacker can exploit the technique to generate a forged individual image without the consent of the image owner. To counter this threat, researchers have developed privacy-protective perturbation techniques that degrade the quality and identity consistency of generated images to mitigate the risk of forgery attacks. Recent studies have employed red-teaming algorithms to attack these protective perturbations to highlight their vulnerabilities. In this work, we theoretically analyze why Blind Face Restoration (BFR) algorithms are suited for red-teaming privacy-protective perturbations. We also empirically demonstrate that BFR not only achieves stronger attack performance than existing red-teaming methods, but also exhibits greater robustness to PGD setting variations and significantly higher computational efficiency compared to purification-based approaches. We further propose a protective perturbation simulation scheme that leverages the trainability of BFR models to enhance attack performance under challenging PGD settings, thereby highlighting the advantage of the trainable BFR models over existing non-trainable red-teaming methods. We conduct adaptive tests in which the protector adjusts the perturbation to defend against BFR. Results show that BFR remains effective even under such defenses. These findings reveal BFR as a significant yet underexplored privacy threat to individual image generation services on the Web.

CCS Concepts

• **Security and privacy** → *Web application security*.

Keywords

Red-teaming Protective Perturbation; Blind Face Restoration; Perturbation Simulation; Adaptive Protection

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1 Introduction

The individual image generation technologies enable users to produce AI-generated portraits of specific individuals by providing personal photos [8, 14, 27]. Such technologies have been broadly integrated into creative web platforms such as [MidJourney](#), [CivitAI](#), [Getty Images](#), and [Exactly.ai](#), making AI-based portrait generation service easily accessible to the public through convenient web interfaces. While these services were originally designed for creative or entertainment purposes (e.g., personalized portrait editing, virtual identity creation), their deployment on web platforms also introduces new risks to privacy and portrait rights [6].

As shown in Figure 1, a forgery attacker can scrape publicly available photos from social media, professional profiles, or any other websites where individuals share personal images. They can use individual image generation techniques to create forged images for malicious purposes, such as identity theft, impersonation, or misinformation campaigns, which pose tangible threats to online identity security. To mitigate this risk, researchers have proposed injecting privacy-protective perturbations into images before they are shared online [16]. As shown in the Figure 1, when an image contains such perturbations, the quality of images generated using individual image generation techniques significantly degrades, and the identity consistency with the image owner is notably reduced.

However, recent studies have shown that red-teaming algorithms can circumvent such privacy-protective perturbations [37]. As illustrated in Figure 1, when red-teaming algorithms are not applied (i.e., w/o Attack), the generated image exhibits poor quality. In contrast,



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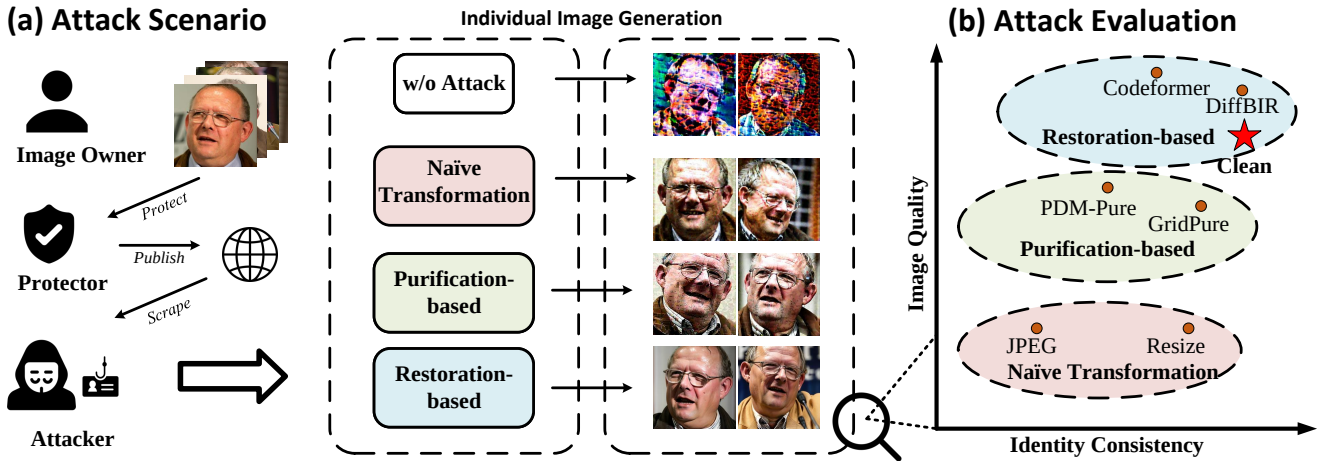


Figure 1: Overview of red-teaming the protective perturbations. (a) The image owner employs a trusted protector to embed privacy-protective perturbations before sharing the images publicly. A malicious actor can then scrape these protected images from online platforms and attempt individual image generation with different attack algorithms to forge the original identity. (b) The effectiveness of each attack method is assessed based on the image quality and identity consistency of the generated outputs with the original image. Restoration-based methods generally yield higher image quality, with DiffBIR achieving the best identity preservation among all evaluated approaches. For comparison, we also report the performance of individual image generation based on unprotected images, denoted as “Clean” in the figures.

when red-teaming algorithms are applied (e.g., naïve transformations, purification-based attacks, or restoration-based attacks), the image quality improves to varying degrees, indicating different levels of degradation in protection performance. Existing attack approaches can be categorized into two types: (1) naïve transformation: adopting hand-crafted transformations like JPEG Compression to transform the image, which often cause significant degradation in image quality; and (2) purification-based methods: leveraging a diffusion model to purify images, which struggle to maintain both high image quality and identity consistency. In Figure 1(b), we also present the performance of individual image generation using clean (i.e., unprotected) images for reference (denoted as “Clean” with a star icon). Compared to this reference, existing attack methods still fall short, indicating they cannot fully bypass the protection.

In this work, we expose and analyze a previously under-explored yet impactful threat to privacy protection in individual image generation: **restoration-based attacks using Blind Face Restoration (BFR) algorithms**. BFR was originally designed for benign tasks such as old photo restoration and facial beautification [17]. We are the first to systematically analyze its potential as a red-teaming algorithm, showing that it significantly outperforms existing attack methods in invalidating privacy-protective perturbations. To better understand how red-teaming transformations affect attack outcomes, we formally model the red-teaming task via an objective function in Section 4. The objective function reveals that the essence of red-teaming lies in reducing the diffusion loss of the protected image while preserving the image quality and identity consistency of the attacked image. However, existing approaches struggle to maintain this balance. A small diffusion loss may indicate that the protective perturbation has been largely neutralized, but this often comes at the cost of substantial information loss,

resulting in poor identity consistency (e.g., PDM-Pure). Conversely, retaining sufficient image information can make the purification incomplete, leaving residual perturbations that lead to high diffusion loss. These residuals interfere with the generation process and ultimately degrade image quality (e.g., GridPure). In contrast, the optimization objective of BFR inherently aligns with the goal of reducing diffusion loss while preserving identity consistency, making it well-suited for attacking protective perturbations. Leveraging powerful generative priors, BFR transforms protected images using high-quality image features, ensuring high image quality after transformation [30]. As a result, advanced BFR algorithms can effectively invalidate the protective perturbation while maintaining both identity consistency and image quality, as shown in Figure 1.

We conduct extensive experiments demonstrating that BFR algorithms outperform existing solutions in attack performance across CelebA-HQ [11] and VGGFace2 [3]. They also exhibit greater robustness to variations in PGD settings and offer higher computational efficiency compared to purification-based methods. We propose a protective perturbation simulation scheme that leverages the trainability of BFR models, showing that their attack performance can be further enhanced under challenging PGD settings. We also perform an adaptive test in which the protector adjusts the perturbation strategy in response to BFR attacks. Results show that even under such adaptive defenses, it remains difficult to counter BFR attacks effectively. These findings highlight the vulnerability of existing protective perturbation schemes to BFR attacks and underscore the need for stronger privacy protection mechanisms for individual image generation services deployed on the Web. The contributions of this work can be summarized as:

- (1) To the best of our knowledge, we are the first to both theoretically and empirically identify Blind Face Restoration

(BFR) methods as effective attack strategies against privacy-protective perturbations.

- (2) We systematically evaluate the defense performance of state-of-the-art protective perturbation methods against advanced attack algorithms. Our results show that the BFR algorithms consistently outperform existing red-teaming approaches across attack effectiveness, computational efficiency, and robustness to varying PGD settings.
- (3) We propose a protective perturbation simulation scheme that leverages the trainability of BFR models to enhance attack performance under challenging PGD settings, demonstrating that trainable BFR models exhibit a clear advantage over existing non-trainable red-teaming methods. We also demonstrate that BFR remains highly effective even under adaptive defense scenarios, underscoring its practical threat to online privacy in individual image generation services.

2 Related Work

2.1 Protective Perturbation

Due to the potential misuse of individual image generation techniques [8, 14, 27] for identity forgery, protective perturbations have been proposed to prevent unauthorized generation [16, 19, 20, 23–25, 28, 31]. AdvDM [20] is the first protective perturbation algorithm to use Projected Gradient Descent (PGD) [25] to update the perturbation. AdvDM updates the perturbation to maximize diffusion loss, ultimately reducing generation quality and preventing unauthorized generation. Subsequent work has built upon this idea, improving the calculation of diffusion loss to enhance protection performance. Some approaches propose updating both the perturbation and the diffusion model parameters to better simulate real-world personalized generation scenarios (e.g., Anti-DreamBooth [16], MetaCloak [24], and CAAT [31]), while others adaptively select time steps when computing diffusion loss (e.g., DisDiff [23], SimAC [28]). However, these methods often fail to discuss their performance when confronted with advanced red-teaming approaches.

Recent red-teaming approaches have explored whether protective perturbations can be invalidated. Hönig et al. [10] reported that simple transformations like resizing and JPEG compression can partially mitigate the protective effects of the perturbations. Zhao et al. [37] proposed a more advanced diffusion-based purification method: GridPure, where the diffusion process is used to add noise that disrupts the perturbation, followed by the reverse diffusion process to reconstruct the clean image, removing the perturbation. PDM-Pure [32] removes perturbations by applying 88× downsampling, and reconstructs the original image using two diffusion models for upsampling. In this work, we assess how advanced protection methods perform against state-of-the-art red-teaming algorithms and show that existing protective perturbations consistently fail to withstand BFR attacks.

2.2 Blind Face Restoration

Blind face restoration (BFR) models aim to learn a mapping from low-quality faces to their high-quality counterparts [18, 22, 26, 30, 33, 34, 38]. However, if we directly train a model to learn the mapping, it will suffer from the over-smoothing problem due to the loss

of fine-grained details in low-quality images [26]. To mitigate this issue, researchers have introduced pretrained generative models into the restoration pipeline. These models serve as “generative prior”, offering realistic facial features that enhance the visual quality of the restored images. PULSE [26] pioneered the use of generative prior for high-quality face restoration. Subsequent methods like GFPGAN [30], built on StyleGAN [12], and CodeFormer [38], which leverages VQGAN [7]. With the growing popularity of diffusion models [9], methods such as DiFace [34] and PGDiff [33] (based on pixel-level diffusion), as well as DiffBIR [22] (built on Stable Diffusion), have demonstrated impressive performance. In this work, we argue that BFR is closely connected to the privacy-protective perturbation red-teaming. Two representative BFR methods, CodeFormer (GAN-based generative prior) and DiffBIR (diffusion-based generative prior), are utilized to conduct attack experiments.

3 Threat Model and Experiment Settings

This section describes the red-teaming scenario for attacking protective perturbations and outlines how it is simulated in our experiments. The setting involves three roles: the **image owner**, the **attacker**, and the **protector**. The attacker obtains the image owner’s photos without authorization and uses individual image generation techniques to synthesize forged images. The protector embeds the protective perturbation into the images to degrade both the visual quality and identity consistency of images generated by the attacker, thereby safeguarding privacy.

Given that protective perturbations are derived from the image to be protected, we assume the protector can apply them to the clean image with authorization from the image owner [16]. In contrast, the attacker has access only to the protected (perturbed) images and must first neutralize the perturbation before applying individual image generation techniques. The goal of the attack is to neutralize the protective perturbations, enabling high-quality image generation unaffected by the perturbation.

Image Owner. Following [16], we sample images from two datasets: CelebA-HQ [11] and VGGFace2 [3]. Based on the identity annotations, we randomly select 50 individuals as image owners and sample four images per individual. These 200 images (50×4) serve as the targets for both protection and subsequent attack.

Protector. We select six state-of-the-art protective perturbation algorithms for evaluation: AdvDM, Anti-DreamBooth, CAAT, SimAC, DisDiff, and MetaCloak. Due to their limited recency, we include AdvDM and Anti-DreamBooth results in Appendix B.2. We use the official implementations released by the authors of each method.

Attacker. Three types of red-teaming algorithms are adopted to launch the attack: (1) Naïve Transformation: JPEG (quality=50), Resize (4× downsampling). (2) Purification-based: GridPure, PDM-Pure. (3) Restoration-based: CodeFormer, DiffBIR.

Personalized Generation. After performing the attack, DreamBooth is adopted to do individual image generation by the prompt “a photo of *sks* person”. For each individual, we generate 16 images. We also provide the results for another individual image generation technique, CustomDiffusion [15] in Appendix B.1.

Metrics. To evaluate attack performance, we assess both the visual quality of the generated images and their identity consistency with

the image owner. For image quality assessment, we adopt CLIP-IQAC [24], an adaptation of CLIP-IQA [29] designed for evaluating the quality of human facial images. We also use LIQE [36], a perceptual quality metric applicable to both human and non-human images. For identity consistency, we use Identity Score Matching (ISM) and Face Detection Failure Rate (FDFR)[16]. ISM computes the cosine similarity between the ArcFace[5] embedding of a generated image and the average identity embedding from the original image set of the same individual. FDFR measures the failure rate of face detection using RetinaFace [4].

4 Analysis of BFR as a Red-Teaming Strategy

In this section, we introduce the objective functions for individual image generation and protective perturbation. We conduct an observational study to formulate a red-teaming objective function. Finally, we justify that BFR algorithms can be taken as a red-teaming algorithm based on the formulated red-teaming objective function. **Individual image generation.** The individual image generation is achieved by finetuning a diffusion model parameterized by θ using diffusion loss $\mathcal{L}_{DM}$ [27]:

$$\min_{\theta} \mathbb{E}_{x \sim X, \epsilon \sim \mathcal{N}(0,1), t} [\mathcal{L}_{DM}], \quad (1)$$

where $\mathcal{L}_{DM} := \|\epsilon_{\theta}(x, c, t) - \epsilon\|_2^2$.

c is the conditioning vector obtained from a text prompt (e.g., “a photo of *sks* person”), t is a discrete time step sampled from the range $[0, 1000]$, and X is a set of images from the same individual. After fine-tuning, the diffusion model can generate a set of samples Y that align with the distribution of X , i.e., Y approximates X in identity. This allows users to generate images of the person in X , and customize the outputs via text. However, this capability raises concerns about privacy and the portrait rights of the person in X , particularly when Y is generated without authorization.

Protective perturbation. The protective perturbation technique is proposed to prevent unauthorized personalized generation [16]. It adds an adversarial perturbation δ to the original image x , producing a perturbed image $\hat{x} = x + \delta$. The perturbation δ is optimized by maximizing the diffusion loss:

$$\max_{\delta} \mathbb{E}_{\hat{x}, \epsilon, t} [\|\epsilon_{\theta}(\hat{x}, c, t) - \epsilon\|_2^2]. \quad (2)$$

The Y generated by $\hat{x}$ will exhibit a lower image quality and reduced identity consistency with X . Thus Y are more likely to be perceived as forgeries. Consequently, protective perturbation serves as an effective method for safeguarding privacy and portrait rights.

Red-teaming objective. The red-teaming algorithm is essentially a transformation function $F_{\phi}(\cdot)$ applied to the protected image $\hat{x}$, which we refer to as an “attack”. This transformation produces a new image $F_{\phi}(\hat{x})$, from which the generated output Y can maintain high visual quality and strong identity consistency with the original image set X . Unlike perturbation generation, which has an explicit objective function, red-teaming applies a fixed transformation F_{ϕ} that does not adapt to the specific input image or protection method. As a result, there is no formal objective function for F_{ϕ} . Due to the lack of a formal objective function, we conduct an observational study to construct a surrogate objective that links the transformation F_{ϕ} to the attack (i.e., red-teaming) performance.

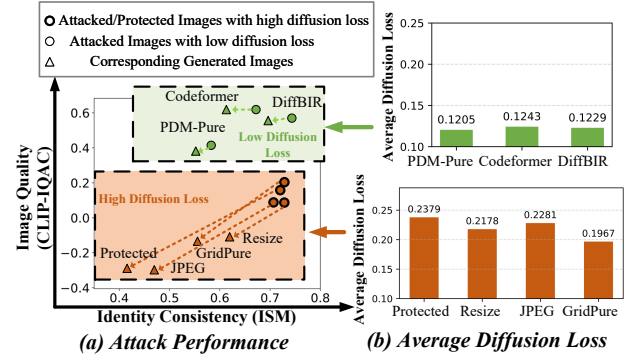


Figure 2: (a) The image quality and identity consistency vary after generation (i.e., from circle to triangle). The top-right corner indicates the optimal attack. (b) The visualization of average diffusion loss for each image group. We use orange to denote high diffusion loss and green for low diffusion loss. With higher diffusion loss, the images suffer from a larger decrement in image quality and identity consistency after generation, indicating a worse attack.

Following the setup in Section 3, we sample 200 images from CelebA-HQ and apply MetaCloak to get the protected images. Various red-teaming algorithms are then applied to the protected image to obtain the attacked images. Finally, personalized generation is performed on both the protected and attacked images to produce the corresponding generated images. We evaluate attack performance based on the identity consistency and image quality of the generated outputs. To assess the impact of transformation, we also compute the identity consistency and image quality of the protected and attacked images and measure their diffusion loss. The diffusion loss is calculated by the average diffusion loss over 1000 randomly sampled time steps.

In Figure 2(a), we compare image quality and identity consistency, where the top-right corner indicates optimal results. Although the protected and attacked images originate from the same images, they differ in quality and consistency (see circles), reflecting the varying effects of red-teaming transformations. After generation (from circles to triangles), the generated images exhibit varying levels of degradation in identity consistency and image quality. Figure 2(b) reveals that lower diffusion loss generally corresponds to less degradation, suggesting that successful attacks should exhibit low diffusion loss. However, diffusion loss alone does not determine the final result. For instance, PDM-Pure reduces diffusion loss but impairs the identity consistency of the attacked images, limiting the identity consistency of the generated output. In contrast, BFR algorithms (e.g., CodeFormer and DiffBIR) not only lower diffusion loss but also better retain image quality and identity consistency of the attacked images, achieving the most effective attacks.

Based on this observation, we hypothesize that three factors govern red-teaming performance (i.e., the triangles’ positions): (1) the identity consistency and (2) image quality of the attacked image, which determine its initial position (see the circles); (3) the diffusion loss of the transformed image, which influences the extent of degradation due to the generation process (i.e., the variation from

circle to triangle). Thus we formulate the red-teaming objective as:

$$\min_{\phi} \mathbb{E}_{x \sim X} \left[\underbrace{\mathbb{E}_{\epsilon, t} [\|\epsilon_{\theta}(F_{\phi}(\hat{x}), c, t) - \epsilon\|_2^2]}_{\text{Diffusion Loss}} + \underbrace{\lambda_{id} \|ID(F_{\phi}(\hat{x})) - ID(x)\|}_{\text{Identity Consistency}} - \underbrace{\lambda_{iq} \|IQ(F_{\phi}(\hat{x}))\|}_{\text{Image Quality}} \right], \quad (3)$$

where the $ID(\cdot)$ and $IQ(\cdot)$ denote the identity information and image quality, respectively.

Connection to Blind Face Restoration. Blind Face Restoration (BFR) is the process of recovering a high-quality facial image x from its degraded version. The degradation occurs due to an unknown image transformation in the real world (e.g., motion blur, sensor noise). The optimization objective of the restoration model $f_{\Theta}(\cdot)$ is:

$$\min_{\Theta} \mathbb{E}_{x \sim X} [\|f_{\Theta}(\Phi(x)) - x\|], \quad (4)$$

where $\Phi(\cdot)$ refers to an unknown degradation. As the degradation function $\Phi(\cdot)$ is unknown, it is approximated using synthetic operators such as Gaussian blur, resizing, additive Gaussian noise, and JPEG compression [17]:

$$\hat{\Phi}(x) = \left((x \otimes k) \downarrow_r + n_{\sigma} \right)_{\text{JPEG}_q}, \quad (5)$$

where the factors k, r, σ, q are randomly sampled. $\hat{\Phi}(\cdot)$ is used to simulate the potential unknown degradation $\Phi(\cdot)$.

The optimization of the BFR model f_{Θ} leads to the reduction of diffusion loss and increasing identity consistency in Equation 3. As the protective perturbation is essentially an additive noise [20], we can interpret the perturbed image $\hat{x} = x + \delta$ as the result of a degradation process $\Phi(x)$. The optimization of the restoration model (Equation 4) aims to approximate $f_{\Theta}(\Phi(x)) \approx x$, which implies that $f_{\Theta}(\hat{x}) \approx x$. We formulate the approximation as follows:

$$\lim_{f_{\Theta}(\hat{x}) \rightarrow x} \left(\mathbb{E}_x \left[\mathbb{E}_{\epsilon, t} [\|\epsilon_{\theta}(f_{\Theta}(\hat{x}), c, t) - \epsilon\|_2^2] + \lambda_{id} \|ID(f_{\Theta}(\hat{x})) - ID(x)\| \right] \right) = \mathbb{E}_{x, \epsilon, t} [\|\epsilon_{\theta}(x, c, t) - \epsilon\|_2^2] \quad (6)$$

$$\leq \mathbb{E}_x \left[\mathbb{E}_{\epsilon, t} [\|\epsilon_{\theta}(\hat{x}, c, t) - \epsilon\|_2^2] + \lambda_{id} \|ID(f_{\Theta}(\hat{x})) - ID(x)\| \right],$$

with $\mathbb{E} [\|\epsilon_{\theta}(x, c, t) - \epsilon\|_2^2] \leq \mathbb{E} [\|\epsilon_{\theta}(\hat{x}, c, t) - \epsilon\|_2^2]$ due to Equation 2 and $\|ID(f_{\Theta}(\hat{x})) - ID(x)\| \geq 0$. As a result, f_{Θ} meets the objective of red-teaming in decreasing diffusion loss and improving identity consistency.

The BFR model guarantees the transformed image quality by leveraging the generative prior. Part of the BFR model weights are borrowed from a pre-trained generative model f_G . These weights are frozen during the training of the restoration model. So we divide the weights into two groups (i.e., $\Theta = \{\psi, G\}$ and $f_{\Theta}(\cdot) = f_G(f_{\psi}(\cdot))$). Let $\mathcal{I}_{\text{high}}$ denote the manifold of high-quality images. The pre-trained generator f_G is optimized to produce samples within this manifold, i.e., $\text{range}(f_G) \subseteq \mathcal{I}_{\text{high}}$. As the restoration model is defined as $f_{\Theta}(\cdot) = f_G(f_{\psi}(\cdot))$, it follows that:

$$\text{range}(f_{\Theta}) \subseteq \text{range}(f_G) \subseteq \mathcal{I}_{\text{high}}, \quad (7)$$

thereby implicitly constraining the restoration model outputs to reside within the high-quality image domain [26, 30]. As a result, f_{Θ} can meet the red-teaming objective in keeping good image quality.

Optimizing the BFR model reduces diffusion loss while improving identity consistency (Equation 6). The generative prior incorporated into the BFR model ensures the high quality of the attacked image (Equation 7). Consequently, BFR model $f_{\Theta}(\cdot)$ can be seen as an effective attack $F_{\phi}(\cdot)$ against protective perturbations.

5 Protective Perturbation Simulation for Enhancing BFR Attack

One key distinction between restoration attacks and purification attacks lies in trainability. While purification methods rely on fixed diffusion models, the restoration-based attacks leverage a trainable BFR model that can learn the inverse of the degradation process (Equation 5). The perturbation itself can be regarded as a special form of image degradation, so the BFR model can learn to remove it through training. However, training a restoration model requires many protected-original face pairs, which in turn requires generating large numbers of protective perturbations. But generating perturbations is costly. As discussed in Section 2.1, existing protective perturbation algorithms share the common strategy of maximizing diffusion loss to decrease image generation quality, which requires a stochastic diffusion denoising step for each PGD iteration. It requires repeatedly running the diffusion model, which results in computational overhead.

To reduce this cost, we propose the Protective Perturbation Simulation Scheme, which uses diffusion-generated noisy images to compute a surrogate diffusion loss. As illustrated in Algorithm 1, we first construct a pool of noisy, low-quality candidates generated by a diffusion model. Given a clean face image x , we apply PGD to obtain a perturbation δ that minimizes a latent-space discrepancy between $x + \delta$ and a sampled noisy candidate. It works because the noisy images have high diffusion loss. By encouraging perturbations that move the original images toward these noisy images, we can simulate the effect of increasing diffusion loss without invoking the denoising process of a diffusion model.

Our simulation scheme eliminates the diffusion step and directly optimizes the perturbed image toward candidate low-quality images, thereby achieving higher computational efficiency while preserving consistency with the protection objective. A runtime comparison is provided in the Appendix B.3.

6 Experiments

6.1 Attack Performance Comparison

As outlined in Section 3, CelebA-HQ and VGGFace2 images are used for individual image generation under different protective perturbations and attack settings. We evaluate the generated results after the attack. In addition, generation from clean (unprotected) images is included as a baseline, reflecting the ideal quality when no protection is applied.

As shown in Figure 9, restoration-based attack algorithms (e.g., Codeformer, DiffBIR) perform significantly better than existing attack methods. From the image quality perspective, naïve transformation methods perform the worst among all evaluated approaches. Purification-based methods achieve relatively better image quality, yet still fall short of gaining image quality as good as using the clean image. In contrast, the restoration-based attack surpasses the performance obtained using the clean image (as indicated by the

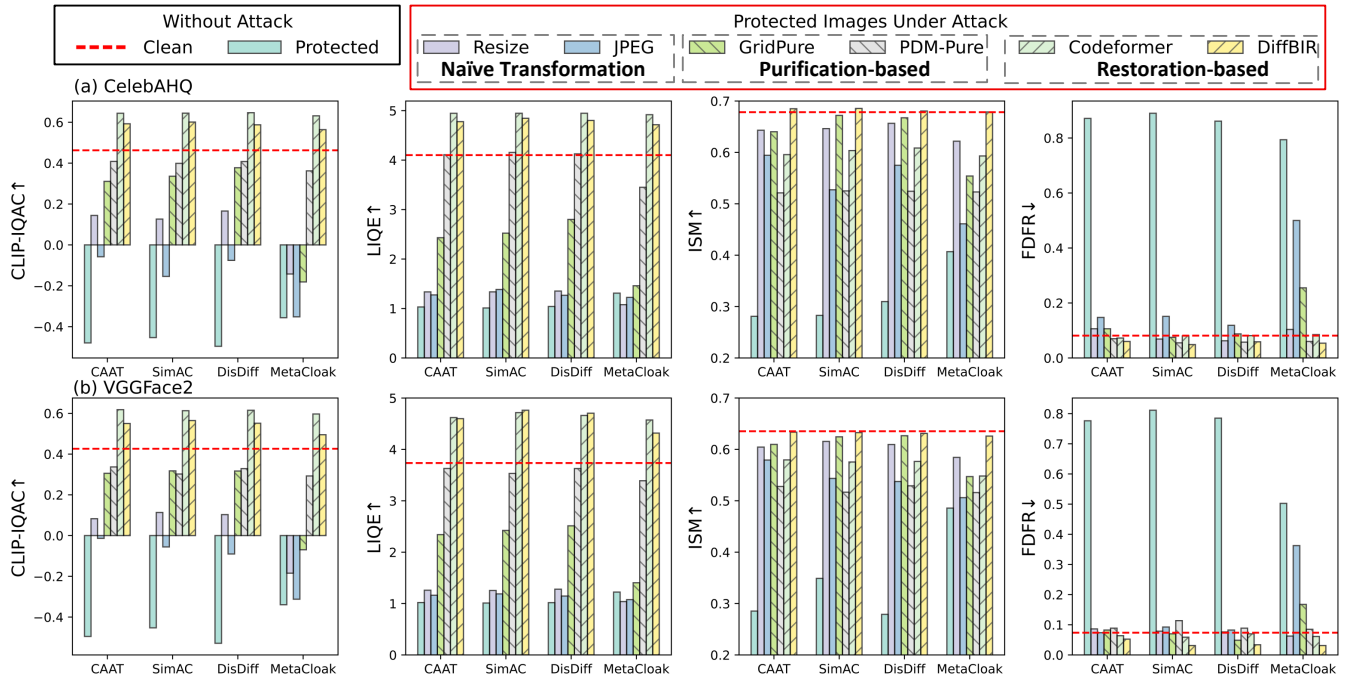


Figure 3: Attack performance comparison of different red-teaming algorithms on different protective perturbations. Each subplot compares performance using different image quality and identity-related metrics (Y-axis). The red dashed line represents the generation quality when the clean image is used to do generation. For each cluster of bars, the leftmost bar (“Protected”) corresponds to the image after applying the protective perturbation at the respective X-axis position. The remaining bars represent the generation quality after attacking the protected images using different red-teaming algorithms.

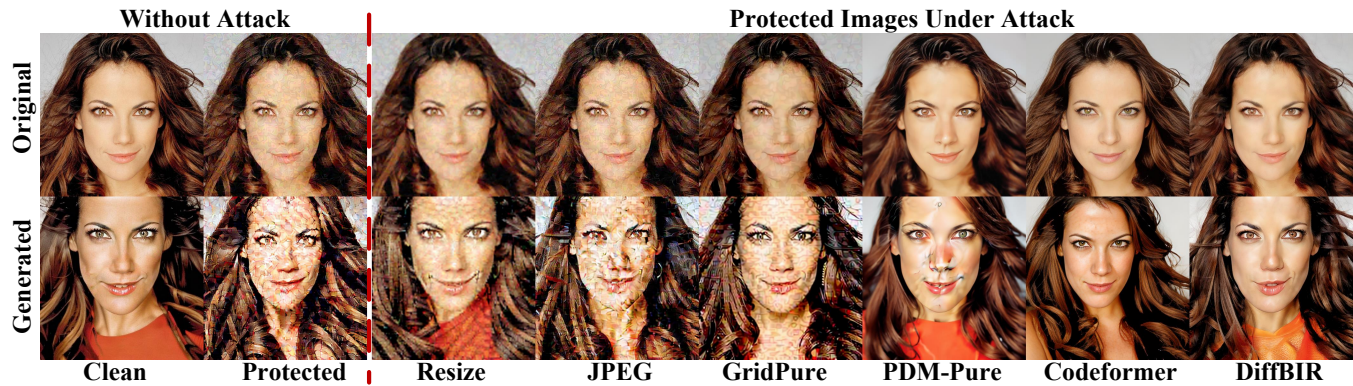


Figure 4: Qualitative comparison of image generation. The “Original” row presents the images used for generation, while the “Generated” row shows the generated results. BFR-attacked images lead to high-quality generation.

red dashed line in the figure). Regarding identity consistency, the restoration-based attack DiffBIR achieves the best performance, with ISM scores comparable to clean images and lower FDFR scores. Considering both image quality and identity preservation, DiffBIR stands out as the most effective attack algorithm.

We also show visual results of MetaCloak-protected images attacked by different algorithms in Figure 4. Naïve transformations

and GridPure struggle to fully eliminate the protective perturbations, leading to visible artifacts in the generated images. PDM-Pure removes the perturbations but compromises identity information, resulting in overly smoothed images with artifacts. The results reveal that purification-based methods fail to balance perturbation removal with identity preservation, whereas restoration-based attacks offer a better trade-off between weakening protection and preserving image quality and identity consistency.

Algorithm 1: Protective Perturbation Simulation Scheme

Input: Stable Diffusion model SD; autoencoder encoder $E(\cdot)$ from SD; face dataset $\mathcal{D} = \{x_i\}_{i=1}^N$; PGD radius r , step size α , iterations T ; number of noisy images M .

Output: Perturbed faces $\{\tilde{x}_i\}_{i=1}^N$ aligned with noisy targets.

Require: Projection operator $\Pi_{x,r}^\infty$ onto the L_∞ -ball around x ; loss $\mathcal{L}(u, y) = \|E(u) - E(y)\|_2^2$.

Step A: Generate noisy targets

$\mathcal{Y} \leftarrow \{\text{SD}(\text{"random noise", random seed})\}_{j=1}^M$.

Step B: Random matching + PGD

foreach $x_i \in \mathcal{D}$ do

 Sample $y \sim \text{Uniform}(\mathcal{Y})$; $x^{(0)} \leftarrow x_i$;

 for $t = 0$ to $T - 1$ do

$x^{(t+1)} \leftarrow \Pi_{x_i,r}^\infty(x^{(t)} - \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(x^{(t)}, y)))$;

$\tilde{x}_i \leftarrow x^{(T)}$;

return $\{\tilde{x}_i\}_{i=1}^N$.

Method	Mean (s)	Variance	Speedup (vs. GridPure)
GridPure	166.461	7.22×10^{-3}	1×
PDM-Pure	31.324	9.91×10^{-3}	5.3×
CodeFormer	0.016	6.14×10^{-3}	10,400×
DiffBIR	5.113	1.32×10^{-3}	32.5×

Table 1: Inference time comparison of different red-teaming algorithms. Bold indicates the lowest time-cost.

6.2 Computation Efficiency Comparison

We sample 30 images to test the inference time of each attack algorithm. As shown in Table 1, restoration-based attack algorithms exhibit better computation efficiency than purification-based methods. CodeFormer exhibits the fastest inference speed among all evaluated methods. While DiffBIR operates at a slower pace than CodeFormer, it still demonstrates approximately 33-fold speedup compared to GridPure. Purification-based attacks suffer from high computational costs due to their design. GridPure requires 1800 time forward passes of a pixel-level diffusion model at a resolution of 256×256 to complete the purification process. PDM-Pure employs two cascaded pixel-level diffusion models at 64×64 and 256×256 scales, each requiring 50 forward steps. In contrast, CodeFormer requires only one forward pass on a pre-trained VQGAN-based restoration model at 512×512 resolution. DiffBIR requires 100 forward passes of a latent diffusion model at 64×64 scale.

6.3 Attack Robustness Comparison

The protective perturbation algorithms are all based on Projected Gradient Descent (PGD), and their performance is influenced by settings such as the PGD radius and step size. A larger radius and step size lead to stronger protection [32]. In this section, we adopt CAAT as the protection method and vary the PGD settings (i.e., step size/PGD radius from 0.005/0.05 to 0.025/0.25) to evaluate the robustness of red-teaming attacks under different protection strengths. CAAT is selected for its efficiency, making it a practical choice for completing the experiments. In Figure 5, all red-teaming algorithms experience performance degradation as the PGD radius increases. However, restoration-based methods exhibit stronger

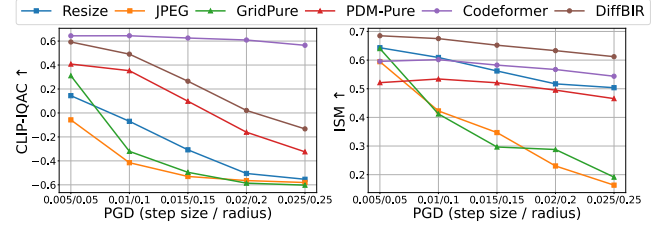


Figure 5: Attack performance of different red-teaming algorithms under varying PGD settings. As the PGD radius and step size increase, overall performance degrades, while restoration-based algorithms exhibit greater robustness.

PGD step size/radius	0.02/0.2		0.025/0.25	
	Non-tuned	Tuned	Non-tuned	Tuned
CLIP-IQAC ↑				
CodeFormer	0.609	0.624	0.565	0.625
DiffBIR	0.021	0.603	-0.133	0.517
ISM ↑				
CodeFormer	0.567	0.570	0.544	0.548
DiffBIR	0.633	0.661	0.612	0.629

Table 2: Comparison of non-tuned vs tuned BFR models with our protective perturbation simulation scheme under large PGD step size/radius.

robustness under these settings. CodeFormer demonstrates the highest robustness in image quality, while DiffBIR maintains the best identity consistency as the PGD radius increases. The strong performance of restoration-based methods can be attributed to their ability to learn to handle complex noise during training (Equation 4), making them more resilient to variations in PGD configurations. Additional visual results are placed in Appendix B.4.

7 Extended Evaluation of BFR

7.1 Fine-tuning with Protective Perturbation Simulation Scheme

The robustness experiments (Figure 5) show that the performance of BFR and other red-teaming methods degrades as the PGD radius or step size increases. However, in this section, we show that by leveraging the trainability of BFR models through our protective perturbation simulation scheme, the BFR attack performance can be further strengthened even under challenging PGD settings.

Following our simulation scheme, we generate perturbed images and fine-tune the BFR models accordingly (The simulation and finetuning details can be found in Appendix A.2). Then, we repeat the robustness experiments under challenging PGD settings. As shown in Table 2, fine-tuning with the proposed simulation scheme improves the attack performance of both CodeFormer and DiffBIR. The visual results in Figure 6 show a similar trend: before fine-tuning, CodeFormer exhibits noticeable color bias with unnatural cyan tones, while DiffBIR suffers from severe artifacts. After fine-tuning, both models achieve visibly better image quality.

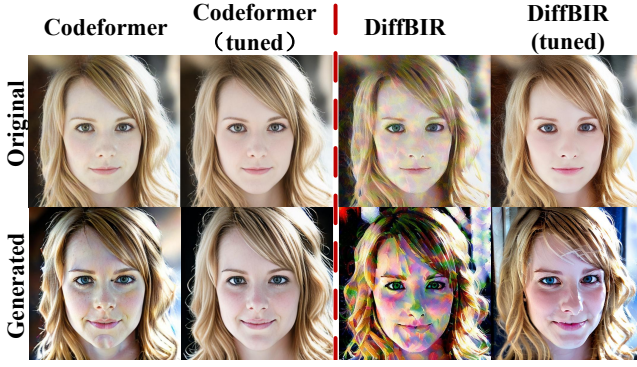


Figure 6: Visual results of the protective image with PGD (0.025/0.25) under attack. The protected image is attacked by non-tuned vs tuned BFR models with our protective perturbation simulation scheme.

Dataset	Metric	No Attack	Attack Scenarios		
			(a)	(b)	(c)
CelebA-HQ	CLIP-IQAC↑	-0.5195	-0.2097	0.0356	0.6220
	LIQE↑	1.0247	1.1912	1.8363	4.9136
	ISM↑	0.2836	0.5702	0.6023	0.5994
	FDFR↓	0.8550	0.0200	0.0425	0.0725
VGGFace2	CLIP-IQAC↑	-0.5652	-0.1999	0.0889	0.5902
	LIQE↑	1.0077	1.0505	1.7697	4.5082
	ISM↑	0.3006	0.5285	0.5978	0.5511
	FDFR↓	0.8150	0.0538	0.0263	0.0613

Table 3: Protection performance of CAAT adapted to DiffBIR under different adaptive test scenarios. The results show a consistent degradation in performance from the “No Attack” setting to adaptive scenarios (a), (b), and (c).

7.2 Adaptive Test

To evaluate the attack’s robustness in a more adversarial setting, we perform an adaptive test in which the protector optimizes the perturbation against a known restoration model, leading to a stronger evaluation of the attack’s robustness. The objective is defined as:

$$\max_{\delta} E_{x \sim X, z \sim \text{Bern}(0.5)} [z \cdot (\|\epsilon_{\theta}(\hat{x}, c, t) - \epsilon\|_2^2) + (1 - z) \cdot (\|\epsilon_{\theta}(f_{\Theta}(\hat{x}), c, t) - \epsilon\|_2^2)]. \quad (8)$$

The protective perturbation is optimized to function after being processed by the restoration model $f_{\Theta}(\cdot)$. CAAT is adapted to generate protective perturbations optimized against the DiffBIR attack. Details on gradient back-propagation through the restoration model for optimization are provided in Appendix A.3. Additional results for CodeFormer attacks are provided in Appendix B.5.

We construct three scenarios for the adaptive test: (a) The protector knows the BFR algorithm and its parameters: Both the protector and the attacker use DiffBIR with identical settings. (b) The protector knows the BFR algorithm (DiffBIR) but not the specific model parameters: The attacker uses a DiffBIR model trained on a different dataset from the one assumed by the protector. (c) The protector is only aware that the attack is restoration-based, without knowledge of the specific algorithm: The attacker employs CodeFormer.

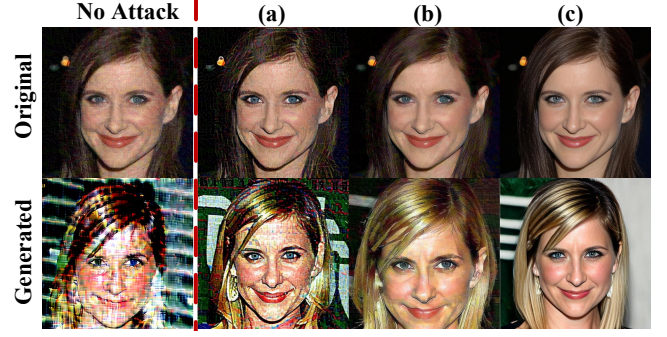


Figure 7: Visual results of the protection. As the attack becomes less predictable to the protector, generated image quality increases, indicating a failure of the protection.

From (a) to (c), the information available to the protector becomes progressively more limited.

We report the protection performance under three adaptive attack scenarios and compare it with the performance when no attack is applied. As shown in Table 3, compared with “No Attack”, applying restoration attacks (i.e., scenario (a), (b), and (c)) significantly degrades the protection effectiveness. Image quality metrics increase consistently from (a) to (c), indicating that protection becomes less effective as the attack becomes less predictable. Identity consistency metrics remain within a range that suggests strong identity preservation across all scenarios, indicating that the protection fails to suppress identity consistency. Figure 7 provides visual examples consistent with the quantitative results. The findings highlight the inherent challenge of defending against restoration-based attacks, especially under limited attacker knowledge as in scenarios (b), (c).

8 Conclusion

This work theoretically and empirically analyzes the threat posed by Blind Face Restoration (BFR) algorithms to privacy-protective perturbations. We show that BFR-based attacks surpass existing red-teaming methods in effectiveness, exhibiting greater robustness to PGD variations and higher computational efficiency than purification-based approaches. We design a protective perturbation simulation scheme that exploits the trainability of BFR models, enabling them to achieve superior attack performance under challenging PGD settings. We also show that under adaptive defenses, BFR attacks remain effective. These findings reveal a critical and previously under-explored vulnerability in current protection mechanisms for online image generation services.

Acknowledgments

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A Implementation Details

We share the [source code](#) of our work.

A.1 Experiment Environment

All experiments were conducted on a server equipped with 4 L40S GPUs (each with 48G) and an Intel(R) Xeon(R) Gold 6426Y CPU. The system had 251 GB of RAM. The software environment included Pytorch 2.4.1 running on Ubuntu 22.04.4 LTS, with CUDA 12.3 and cuDNN 9.1.0.70 for GPU acceleration. We didn't do distributed training, so the experiment can be conducted using one GPU.

A.2 Details on BFR Models' Tuning

For the simulation process, we sample facial images from the FFHQ dataset [13] and generate 200 noisy images. Each face is perturbed for 50 PGD steps to obtain the final simulated protected images used for training. The step size/radius is set to 0.025/0.25. For the training of BFR models, we adopt the official training code. For **CodeFormer**, we fine-tuned the model without the CFT module for 2,000 steps with a learning rate of 2×10^{-5} and a batch size of 4. For **DiffBIR**, we fine-tuned only the SwinIR module for 10,000 steps with a learning rate of 1×10^{-4} and a batch size of 16.

A.3 Details on Adaptive Test

We adopt the optimization strategy from CAAT [31] to update the protective perturbation. The step size and PGD radius are fixed at 8×10^{-3} and 8×10^{-2} , respectively. The updates are performed for 500 steps. Optimization described by Equation 8 can be seen as an extension of the Expectation over Transformation (EoT) paradigm [1] and is compatible with any protective perturbation method. Unlike traditional EoT, which applies simple, input-independent operations (e.g., rotation, blur), Equation 8 incorporates a structured restoration model, requiring consideration of its architecture to enable proper gradient computation. For Codeformer, it adopts a VQGAN [7] architecture and performs restoration by mapping the input image to the vectors in the VQGAN's codebook in a quantization manner. Although the quantization manner is not differentiable, it can be solved by a Straight-Through Estimator[2]. The estimator enables gradient calculation between the restored image and the input image, making the optimization of Equation 8 feasible. For DiffBIR, it adopts a two-stage restoration pipeline. It first adopts a SwinIR [21] architecture to extract the semantic feature of the face. Then, it employs a ControlNet [35] architecture to project the features onto a high-quality face. We observe that the perturbation is primarily removed in the first stage, while the second stage mainly serves to add realistic texture details to the image. Therefore, instead of performing the two-stage restoration to optimize Equation 8, we only perform the first stage restoration.

A.4 Algorithm Hyper-parameters

Protective Perturbation. The random seed is 1042. We use the code from: [AdvDM](#), [Anti-DreamBooth](#), [CAAT](#), [MetaCloak](#), [DisDiff](#), [SimAC](#). We follow the default setting in the repository.

Personalized Generation. The DreamBooth random seed is 10042. We use the code in [DreamBooth](#), [Custom Diffusion](#). We follow the default setting in the code repository.

Purification-based attack. We use the code from: [Gridpure](#), [PDM-Pure](#). We follow the default setting in the code repository.

Restoration-based attack. We use the code from: [Codeformer](#), [DiffBIR](#). We follow the default setting in the code repository.

Evaluation Metrics. [ISM](#), [FDFR](#), [CLIP-IQAC](#), [LIQE](#).

B More Experiment Results

B.1 Other Personalized Generation Technique

We replicate the experiment in Section 6.1 on the CelebA-HQ dataset using Custom Diffusion [15]. As shown in Figure 8, the restoration-based attack again achieves the strongest performance among all attack methods.

B.2 AdvDM and Anti-DreamBooth Performance

We place the performance of Anti-DreamBooth [16] and AdvDM [20] in Figure 9. The restoration-based attack still achieves the strongest performance among all attack methods.

B.3 Time cost Comparison

We perturb a batch of 4 images and report the runtime in Table 4. SimAC incurs additional overhead from step selection, while MetaCloak requires expensive surrogate training. Our method achieves the lowest per-step cost (0.16s) and completes the entire process within 8.09s, resulting in up to **1,635× speedup** over MetaCloak.

B.4 Visual Results for Robustness Experiment

As shown in Figure 12, increasing the PGD radius makes the protective perturbation more visually noticeable. Nevertheless, restoration-based methods exhibit stronger robustness under these challenging settings. CodeFormer consistently removes the perturbation effectively across all settings. Although DiffBIR introduces slight texture artifacts under the (0.025/0.25) setting, it still maintains good identity consistency in terms of facial structure. These findings are also visually supported by the generation results in Figure 13.

Method	PGD Step (s)	Total Time (s)	Speedup vs Ours
SimAC	4.66	338.86	41.9×
DisDiff	4.27	213.48	26.4×
CAAT	0.31	77.33	9.6×
MetaCloak	0.91	13,239.19	1,635.8×
Ours	0.16	8.09	1.0×

Table 4: Time consumption comparison. “PGD Step” denotes the average time per PGD iteration. “Speedup vs Ours” shows the relative acceleration achieved by our method.

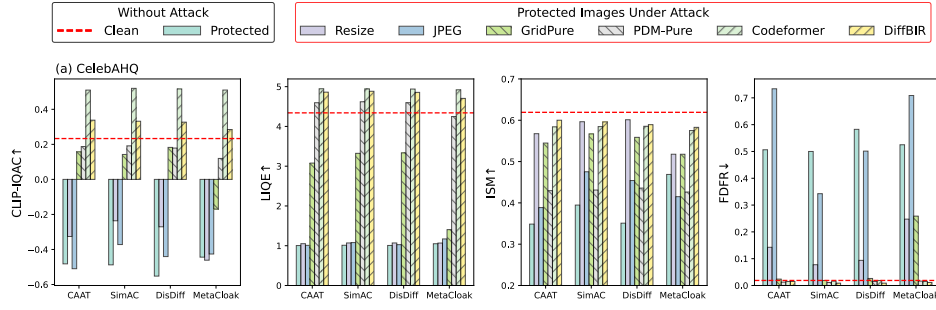


Figure 8: Attack performance comparison of different red-teaming algorithms on different protective perturbations. The generated images are obtained by Custom Diffusion. Each subplot compares performance using different image quality and identity-related metrics (Y-axis). The red dashed line represents the generation quality when the clean image is used to do generation. For each cluster of bars, the leftmost bar (“Protected”) corresponds to the image after applying the protective perturbation at the respective X-axis position. The remaining bars represent the generation quality after attacking the protected images using different red-teaming algorithms.

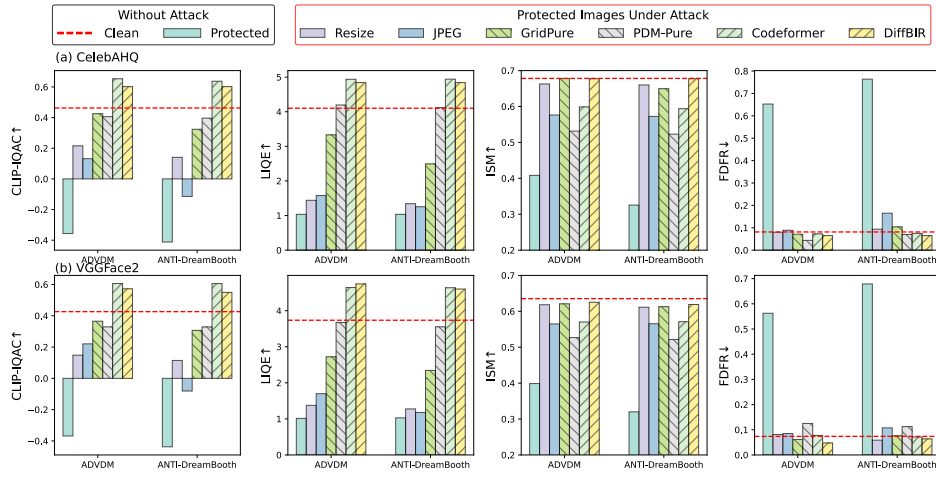


Figure 9: Attack performance comparison of different red-teaming algorithms on Anti-DreamBooth and AdvDM.

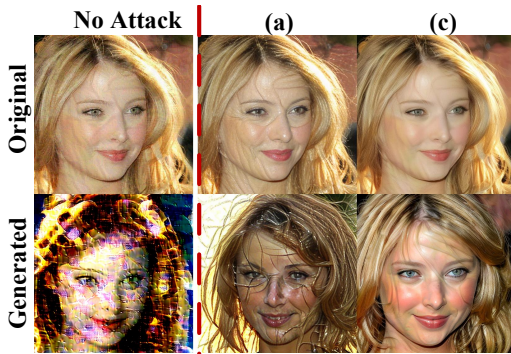


Figure 10: Visual results of the protection. As the attack becomes less predictable to the protector, generated image quality increases, indicating a failure of the protection.

Dataset	Metric	No Attack	Attack Settings		
			(a)	(b)	(c)
CelebA-HQ	CLIP-IQAC↑	-0.5027	0.3400	-	0.3271
	LIQE↑	1.0129	2.8767	-	4.1028
	ISM↑	0.2330	0.5017	-	0.6068
	FDFR↓	0.8988	0.1125	-	0.0688
VGGFace2	CLIP-IQAC↑	-0.5289	0.2536	-	0.3753
	LIQE↑	1.0021	2.4856	-	3.7722
	ISM↑	0.3101	0.4816	-	0.5943
	FDFR↓	0.8438	0.0550	-	0.0438

Table 5: Protection performance of CAAT adapted to Codeformer under different adaptive test settings.

B.5 Adaptive Test for Codeformer

We replicate the experiment from Section 7.2, this time adapting CAAT to use CodeFormer. In scenario (c), DiffBIR is used as the attack model. Since CodeFormer does not provide an official second set of parameters, we omit the results for scenario (b). Nonetheless,

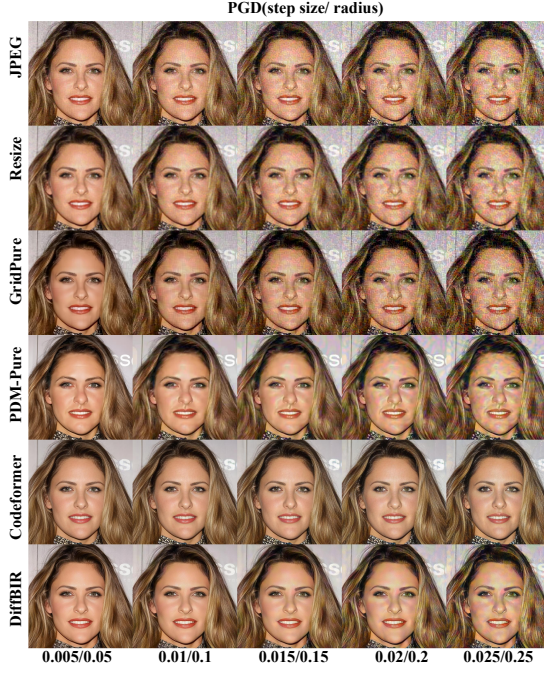


Figure 12: The attacked images of different red-teaming algorithms under different PGD settings.



Figure 13: The generated results based on attacked images produced by different red-teaming algorithms, each evaluated under different PGD settings.

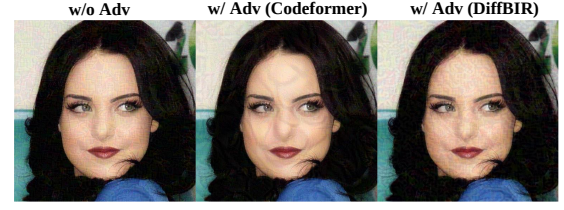


Figure 11: Qualitative comparison of images after applying protective perturbation.

PGD (step size/radius)	CLIP-IQAC $\uparrow$	ISM $\uparrow$
0.01 / 0.1	0.19	0.73
0.015 / 0.15	-0.02	0.68
0.02 / 0.2	-0.20	0.60
0.025 / 0.25	-0.31	0.52

Table 6: As the PGD radius increases, the perturbation becomes more perceptible, leading to a degradation in the visual quality and a loss of identity information.

from (a) to (c), we can still observe a clear degradation in protection performance, as shown in Table 5 and Figure 10.

B.6 Perceptibility Problem of the Protective Perturbation

Although adapting the protective perturbation to the target restoration model or increasing the PGD radius can enhance the effectiveness of protective perturbations, these improvements come at the cost of increased perceptibility. The increased perceptibility of the perturbation makes it more likely to be noticed by users, thereby reducing their willingness to adopt such protection techniques.

High Perceptibility of Adaptive Protective Perturbation. As shown in Figure 11, adapting the perturbation to defend one restoration model makes the perturbation more noticeable, even with the same PGD radius. We also conduct a human study. We randomly selected 20 image groups, each containing (1) the original image, (2) an image with perturbation that is not adapted to a restoration model (w/o Ada), and (3–4) two images with perturbation adapted to DiffBIR or CodeFormer (w/ Ada). 13 volunteers, without knowledge of the perturbation design, were asked to identify the image with the most visible perturbation in each group. Images with w/ Ada (CodeFormer) were chosen in 47% of cases, w/ Ada (DiffBIR) in 52%, and w/o Ada in only 1%.

High Perceptibility of Large PGD radius. As the PGD radius increases, the perturbation becomes more noticeable, leading to a degradation in both visual quality and identity consistency of the protected image. This is evidenced by the decreasing CLIP-IQAC and ISM scores in Table 6, indicating lower image quality and greater identity loss, respectively. These results suggest that while adapting to the target restoration model and increasing the PGD radius can improve protection effectiveness, such gains come at the cost of increased perceptibility. There remains no effective method that achieves both imperceptibility and robustness against restoration-based attacks from BFR algorithms.